

The code for `$a_2$` is:

Please provide a brief explanation of your answer.

No, Jordan's claim is not true.

The code defines a_n as the limit of $\text{Compute } a_{2(n)}$. Analyzing the sums shows

$$a_2 = 1 + 2 \cdot 72^{-10^{24}} - 2 \sum_{k=3}^{\infty} 72^{-10^{23} \cdot k}.$$

The dominant correction term is $-2 \cdot 72^{-3 \cdot 10^{23}}$, which is negative, so $a_2 < 1$ (though extremely close to 1).

With dosage rates $a_1 = 1$ and $a_2 < 1$, minimizing total time $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$ forces using only the faster therapy: the unique minimizer is $(x_1, x_2) = (1, 0)$.

The infinity-norm distance from $(0.3, 0.7)$ to $(1, 0)$ is $\max(0.7, 0.7) = 0.7$, which is **not** ≤ 0.4 .

(The later revelation about $(0.6, 0.4)$ being within 0.45 of a minimizer is true, as its distance to $(1, 0)$ is $0.4 \leq 0.45$.)